Certified Knowledge Editing: Closed-Form Reachability Guarantees for Additive Edits to a Frozen Language Model

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Abstract

Knowledge edits to language models are validated only after the fact. We study the reverse contract: a certificate, computed *before* the edit, of which edits can succeed. HLM5 attaches a slot memory to a frozen decoder-only transformer: a hidden-state query selects slots through a hard cosine gate and adds the value to the pre-head representation. Gate and read see only the unperturbed hidden state, so logits are affine in the edit strength and single-token reachability is one-dimensional linear feasibility: a closed-form certificate, one pass over the head matrix, accepting or refusing each edit in advance. On a frozen 1B trunk, 78.4% of 1,200 single-token targets are reachable at a novel-entity key—79.4% ± 2.2% (mean ± SD) across 60 keys—and readout-level residual synthesis rescues 40/40 sampled unreachable targets (median $\beta = 20$). The certificate also *doses* each edit: a global strength overshoots the certified interval, silently failing reachable edits (efficacy 0.529); the certified $\beta^\star$ flips exactly the certificate-reachable set (0.765); synthesis rescues the rest (1.000)—at off-support locality 1.000, zero gradients, paraphrase transfer 0.020. A gate-matched static logit bias matches the dosed pipeline on every measured axis: the value is the a-priori admission test, certified dose, synthesis rescue, and audit trail, not raw editing power. On 300 CounterFact records, pre-edit certificate difficulty predicts ROME's paraphrase transfer ($p < 0.01$; fine-tuning directionally, $p = 0.02$) and neighborhood damage for ROME and GRACE ($p < 0.01$). Fine-tuning generalizes only by collapsing locality; HLM5 is a zero-gradient, certificate-gated, exact-key editor whose every operation emits a verifiable receipt. Full head-to-head evaluation against MEMIT, MEND, and SERAC on standard benchmarks remains open.

1 Introduction

Post-hoc knowledge editing asks how to install a new fact into a trained model so that the edit (i) takes effect, (ii) does not disturb unrelated behavior, (iii) can be reversed, and (iv) can be audited. Today these properties are established, if at all, only *after* the edit is installed; we instead ask for an *a-priori certificate* that an edit will take and stay local, and treat a failed certificate as a first-class, audited refusal. Fine-tuning and parameter-efficient tuning install facts by gradient descent, mixing them into trained parameters so that locality and reversibility must be re-checked per edit. Retrieval keeps the trunk fixed but installs knowledge in a prompt-assembly layer. Locate-and-edit methods such as ROME and MEMIT [33, 34] mutate trunk weights directly, and deletion is itself another edit. Wrapper editors such as SERAC and GRACE [36, 15] instead attach an external, key-addressed store that defers to the frozen model off-scope.

We place our method, HLM5 (Hopfield Language Model 5), in this last family and ask what becomes provable when the external read is constrained to a particular form. HLM5 attaches a slot memory to a frozen decoder-only trunk; a hidden-state query selects slots through a hard cosine gate, and the selected value is written *additively* into the pre-head representation at a single position. Because the gate and the read depend only on the unperturbed hidden state, the perturbed logits are affine in the scalar edit strength β. Reachability of a single-token target then reduces to one-dimensional linear feasibility, which we solve in closed form. The result is a certificate—computable before injection—that labels each target as reachable or unreachable under the deployed configuration. Editing becomes an admission test with an a-priori guarantee, not a gradient step whose locality is audited after the fact.

HLM5 is also the certified fragment of a broader "energy language" program that treats memory slots as attractor basins to be surveyed, injected, strengthened, forgotten, and transplanted rather than retrained [40, 41]; we consolidate that correspondence, and operational checks of the operator set on a real co-trained memory, in Section 8.2.

Contributions.

1. **A reachability certificate (Section 5).** Because the post-edit logit margins are affine in the edit strength (Lemma 1), we give a closed-form test (Proposition 1) for whether a finite edit strength makes a target win, and a corollary (Corollary 1) that certifies unreachable targets a priori. On the frozen 1B trunk the certificate yields a full operating envelope—259 of 1,200 targets certified unreachable in advance at the reference key, stable at $79.4\% \pm 2.2\%$ reachable across 60 diverse keys—and residual synthesis rescues all 40 sampled unreachable targets in a readout-level verification (Section 7).

2. **A zero-tax co-training result (Section 8.1).** Under a matched-pair protocol, adding the memory module changes validation perplexity by $+0.018\%$ at 136M and -0.116% at 1B.

3. **A governed update contract (Sections 6–8).** Injection, tamper-evident receipts, edit, forget, and an exact off-support fallback are evaluated as first-class behaviors, with measured boundaries reported alongside the successes.

The scope is deliberately narrow. Edits are single-token and exact-key; paraphrase transfer is weak; the additive read degrades multi-query in-context recall; the capacity studies are synthetic; numbers are single-seed; and head-to-head baselines on standard editing benchmarks are not yet run. We state each as a limitation with an explicit pass/fail experiment (Section 9).

2 Related Work

We compare methods by update contract rather than by model family. *Fine-tuning and PEFT*—continued training, LoRA [20], prefix tuning [32], constrained fine-tuning [56]—install facts by optimization. *Retrieval* [28] conditions a frozen generator on retrieved passages, and in-context editing [54] installs the counterfactual directly in the prompt. *Datastore memory*, kNN-LM and RETRO [23, 1], interpolates the output distribution with a nearest-neighbor distribution over a key–value store. *Locate-and-edit* methods—knowledge neurons [4], ROME and MEMIT [33, 34], and their successors PMET [30] and AlphaEdit [10]—solve a constrained update to feed-forward projections, although the localization premise itself has been questioned [16]. *Wrapper and meta-learned editors*—KnowledgeEditor [6], MEND [35], SERAC [36], CaliNet [8], T-Patcher [21], GRACE [15], MELO [52], WISE [47], Larimar [5], and MEMOIR [48]—wrap the frozen trunk with an external cache, codebook, side memory, or hypernetwork. Surveys and benchmarks [51, 53] and toolkits [46] standardize the evaluation around reliability, generalization, locality, and portability [55] on CounterFact [33] and zsRE [27]. A parallel line documents what editing costs: ripple effects on logically related facts [3], knowledge distortion [31], degradation of general abilities [12], and gradual or even catastrophic collapse under sequential weight edits [14, 50]—failure modes that motivate keeping the trunk frozen and the edits external, as here.

Representation-space editing and steering. Closest in mechanism to our additive pre-head write are methods that edit *activations* rather than weights: REMEDI learns fact encodings added to hidden representations [17], activation addition steers generation with difference vectors [45], inference-time intervention shifts attention-head outputs along learned directions [29], and representation engineering generalizes the recipe [57]; the reliability of such steering vectors in and out of distribution is itself under scrutiny [44]. The additive hidden-state write is therefore established prior art. What none of these provide—and what the specific combination of a hard support gate with a write at the *pre-head* position makes possible—is a *closed-form, a-priori reachability certificate*: because our gate and read depend only on the unperturbed hidden state, every logit margin is affine in the edit strength, and admission or refusal of an edit is decidable before it is applied (Section 5).

Associative memory and modern Hopfield networks. Classical Hopfield networks store patterns as attractors of an energy landscape [19]; Dense Associative Memory raises storage capacity by sharpening the interaction to a high-order polynomial [24, 25], its continuous, softmax form underlies modern Hopfield layers [42], and engineered energy functions have since been built into transformer blocks themselves [18]; see Krotov et al. [26] for a current treatment. HLM5's gate (§4) is exactly such a degree-κ associative read, but attached *externally* and coupled *additively* to a frozen trunk rather than realized as the network's own dynamics—which is what makes each edit certifiable (§5).

HLM5 belongs to the wrapper-editor family: GRACE already maintains a key-addressed codebook that defers to the frozen trunk off-scope, and SERAC routes in-scope queries to a counterfactual store. HLM5 claims no novelty for the wrapper architecture itself, nor for the additive activation write (REMEDI, ActAdd, ITI above). Its specific, analyzable choice is an *additive* pre-head read with a hard cosine gate, which makes the logits affine in the edit strength and therefore admits the closed-form reachability certificate of Section 5—to our knowledge the first a-priori admission test with refusal receipts in the editing literature. We position HLM5 within this family and treat the head-to-head benchmark comparison as the primary open evaluation (E1, Section 9).

3 Problem Statement and Preliminaries

Notation. Let f_θ be a decoder-only language model with parameters θ mapping a prefix $x = (x_1, \ldots, x_t)$ to a final hidden state $h = f_\theta(x) \in \mathbb{R}^d$ and, through a tied head $W \in \mathbb{R}^{|V| \times d}$, to logits $z = Wh$ and a distribution $p_\theta(\cdot \mid x) = \mathrm{softmax}(Wh)$. Vectors are columns in inner products; $W_y \in \mathbb{R}^d$ denotes the y-th row of W; $\mathrm{unit}(x) = x/\|x\|_2$; hidden states act as row vectors where the whitening map A multiplies on the right. A *fact* is a key–target pair (x, y) with a single target token $y \in V$. We restrict targets to one token throughout; multi-token targets—sequences of such installs, at the cost of the per-step guarantees below—are deferred to future work (E4, Section 9) [22]. We write $\mathcal{D}_F$ for the facts to install and $\mathcal{D}_H$ for held-out text, the latter taken disjoint from the support of installed keys.

Problem. Given a trunk trained once and facts $\mathcal{D}_F$ arriving afterward, install each (x, y) so that (i) *efficacy*: $\arg\max_{y' \in V} p_{\theta'}(y' \mid x) = y$; (ii) *locality*: behavior on $\mathcal{D}_H$ is preserved; (iii) *reversibility*: an installed fact can be edited or deleted; (iv) *auditability*: every operation emits a verifiable receipt. Criteria (i)–(ii) are the editing literature's reliability and locality/specificity; *generalization* (survival under paraphrases) is out of scope by construction and measured empirically (Section 8.5).

We state the existing families as objectives over $(x, y) \in \mathcal{D}_F$. Supervised fine-tuning installs the fact by

$$\mathcal{L}_{\mathrm{FT}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}_F}[-\log p_\theta(y \mid x)]\,, \tag{1}$$

spreading the update across all of θ. LoRA restricts the write to a low-rank delta on a frozen internal weight $W_0 \in \mathbb{R}^{d\times k}$ (any projection other than the head W),

$$W_0' = W_0 + \tfrac{\alpha}{r}\, BA, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d, k). \tag{2}$$

Retrieval-augmented generation marginalizes over retrieved passages c with the trunk fixed and the corpus $\mathcal{D}_R$ as the editable memory,

$$p(y \mid x) = \sum_{c\in R_\eta(x)} p_\eta(c \mid x)\, p_\theta(y \mid x, c), \tag{3}$$

the RAG-Sequence form of Lewis et al. [28]. kNN-LM interpolates the output with a datastore distribution under a global weight λ,

$$p(y \mid x) = \lambda\, p_{k\mathrm{NN}}(y \mid x) + (1-\lambda)\, p_\theta(y \mid x). \tag{4}$$

ROME edits a feed-forward down-projection W_ℓ, viewed as a linear associative memory, by the rank-one update that installs (k_*, v_*) while least-squares preserving prior associations with uncentered key covariance $C = \mathbb{E}[kk^\top]$,

$$\widehat{W}_\ell = W_\ell + \frac{(v_* - W_\ell k_*)\,(C^{-1}k_*)^\top}{(C^{-1}k_*)^\top k_*}. \tag{5}$$

Across these, the update mutates either trunk parameters (fine-tuning, LoRA, ROME/MEMIT) or input context (RAG, prefix-tuning), and locality, reversibility, and deletion must be established after the fact. HLM5 instead fixes θ and installs each fact as an entry in an external gated memory (Section 4).

4 Method

Frozen trunk. The trunk is a standard decoder-only transformer with tied embedding and head. The 1B configuration uses width 1792, 24 layers, 14 heads, sequence length 1024, and vocabulary 65,536. We use the memory in two distinct ways and keep them separate throughout: the *hybrid arm* co-trains the memory module during pretraining (the no-tax study, Section 8.1); the *injection protocol* attaches an editable memory to the *frozen baseline* trunk at inference (the edit/forget study, Section 8.3). Figure 1 contrasts the two.

Editable memory. The memory holds slots $s_i = (k_i, v_i, \alpha_i, \mathrm{label}_i, \mathrm{active}_i)$. From a hidden state h the query is mean-centered and optionally whitened to counter the anisotropy of trained hidden states [9, 11],

$$q = (h - \mu)A, \tag{6}$$

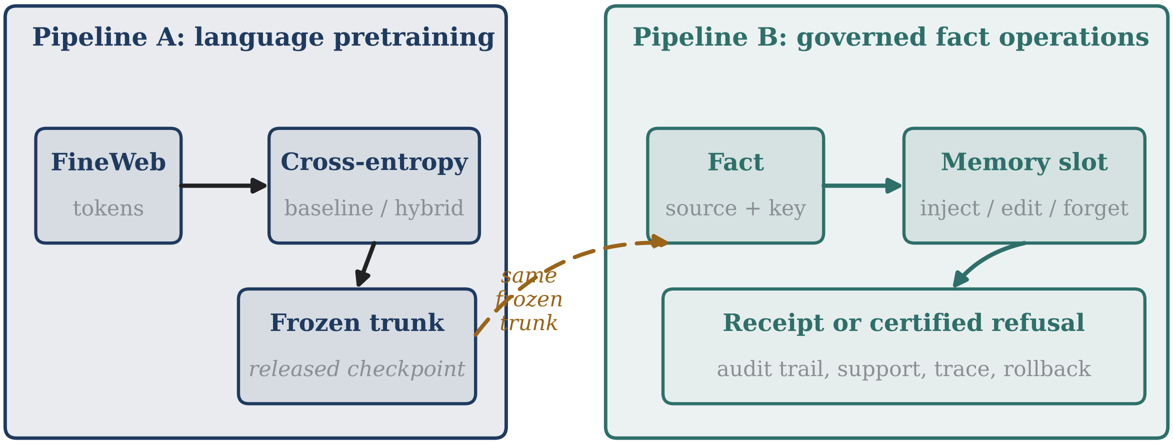

Figure 1: Two pipelines. Pipeline A pretrains the trunk once (optionally co-training the memory module); Pipeline B governs fact create/edit/forget on the *same* frozen trunk, with receipts. Pipeline B never retrains the Pipeline A trunk; no per-fact gradient.

with μ the mean hidden state over a calibration set of fact-key and neutral prompts and A identity or a ZCA whitening matrix estimated from the same calibration set; both are frozen at audit time. With q and keys unit-normalized, slot i scores through a degree-κ cosine kernel—a Dense Associative Memory interaction of polynomial order [24]—gated by an active flag and a nonnegative strength,

$$s_i(q) = \text{active}_i \cdot \text{relu}(\alpha_i) \cdot \max\big(0, \cos(q, k_i)\big)^\kappa, \qquad \kappa = 5, \tag{7}$$

and the read is a temperature-softmax mixture of values,

$$r(q) = \text{softmax}\big(s(q)/T\big)\, V, \qquad T = 0.1. \tag{8}$$

The memory perturbs the pre-head representation—after the trunk's final layer normalization, immediately before the bias-free tied head—and only when the maximum score crosses a hard support threshold τ:

$$g(q) = \begin{cases} \max_i s_i(q) & \max_i s_i(q) \geq \tau \\ 0 & \text{otherwise,} \end{cases} \qquad h' = h + \beta\, g(q)\, r(q), \quad z' = W h', \tag{9}$$

with $\tau = 0.95$ and edit-strength scalar β. We write $\beta_{\text{eff}} = \beta\, g(q)$ and $\bar{r} = r(q)$ for the realized residual. Figure 3 shows the full forward pass.

Value construction. For single-token injection the slot value is the unit output embedding of the target,

$$v = \text{unit}(W_y). \tag{10}$$

A target y is *head-reachable* if its embedding row is its own nearest neighbor in the tied head, $\arg\max_j (W W_y)_j = y$; at 1B, 10.27% of the vocabulary is head-reachable. A head-geometry audit shows this is a property of the output head, not of the memory: reachability varies sharply with embedding norm (Figure 2)—from 30.7% in the lowest-norm decile to 2.7–6.7% in the middle

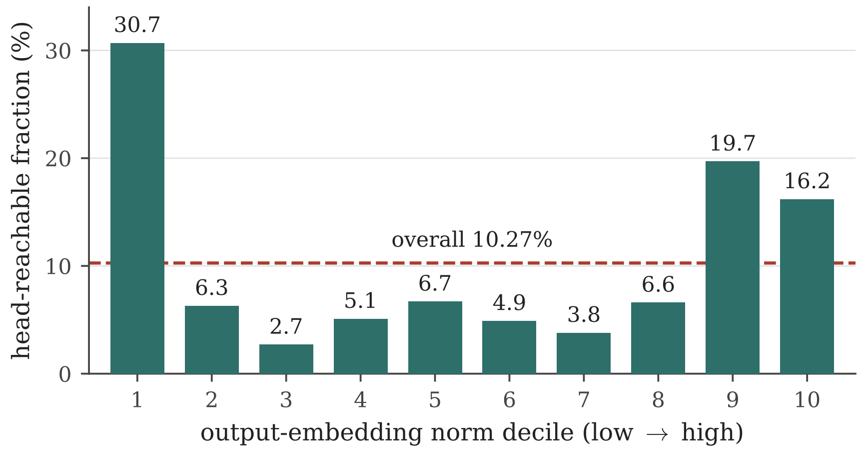

Figure 2: Head-reachability is a property of the output head, not the memory. The head-reachable fraction of the vocabulary varies sharply across output-embedding norm deciles (30.7% in the lowest decile, 2.7–6.7% in the middle, rising again at high norm), around an overall 10.27%; unreachable tokens sit a median of 6,472 ranks from their own nearest neighbor. Measured on the frozen 1B trunk (`headgeom_metrics.json`).

deciles—unreachable tokens have a median self-nearest-neighbor rank of 6,472, and their confusers are dominated by high-frequency tokens and punctuation (e.g. " that" and " with", shown with their leading space, are outranked by ","). This is the "stolen-probability" effect [7] produced by the anisotropic, narrow-cone geometry of tied output embeddings [11, 9] and bounded ultimately by the rank of the softmax head [49]. Many edit failures are therefore head-geometry failures rather than memory failures. Head-reachability is necessary but not sufficient for an edit at a given h; sufficiency is the β-feasibility test of Section 5.

5 Reachability and Off-Support Fallback

All statements below concern a single decoded position with h the unperturbed hidden state, $\bar{r} = r(q)$ the realized residual, and $\beta_{\text{eff}} \geq 0$. The write modifies only the readout at that position—it enters neither the residual stream nor the attention state—so within a forward pass it cannot affect other positions' hidden states; under generation, later positions are affected only through the emitted token, and each per-position guarantee is conditional on its realized prefix. Feasibility is stated in $\beta_{\text{eff}} = \beta\, g(q)$; it transfers to the deployed dial β exactly when the gate fires at the key ($g(q) \geq \tau > 0$), which the measured gate selectivity guarantees at exact keys (§7).

Lemma 1 (Margins are affine in edit strength)**.** *The gate $g(q)$ and read $\bar{r}$ depend only on q, which is a function of the unperturbed h. Hence, for any competitor j, the post-edit margin of target y is affine in β_{eff}:*

$$m_{yj}(\beta_{\text{eff}}) = (W_y - W_j)^\top h' = a_j + \beta_{\text{eff}}\, b_j, \quad a_j = (W_y - W_j)^\top h, \quad b_j = (W_y - W_j)^\top \bar{r}.$$

Proof. By (9), $h' = h + \beta_{\text{eff}}\bar{r}$ with $\bar{r}$ and the gate constant in β_{eff} since they are determined by $q = (h - \mu)A$. Substituting and using linearity of $(W_y - W_j)^\top(\cdot)$ gives the stated affine form. $\square$

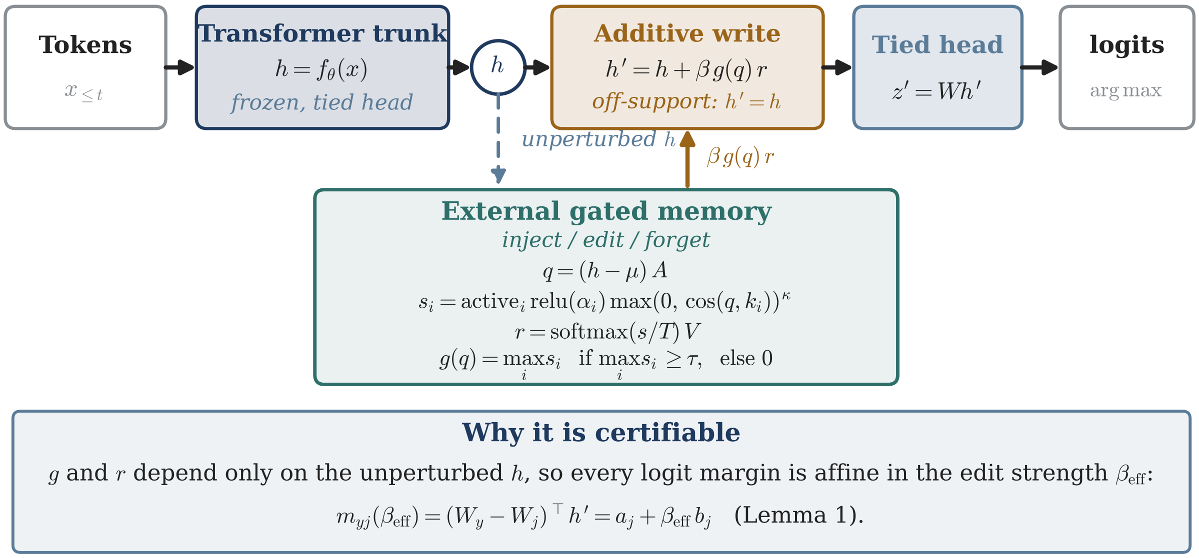

Figure 3: HLM5 forward pass. The frozen trunk produces the hidden state h; a mean-centered, whitened query $q = (h - \mu)A$ scores slots with a degree-κ cosine kernel, and a hard support gate either writes the boosted value $\beta\, g(q)\, r$ additively into the pre-head state or leaves the trunk untouched. Because the query, gate, and read all read the *unperturbed* h, every logit margin is affine in the edit strength (Lemma 1)—the property the certificate rests on.

Proposition 1 (β-feasibility)**.** *Target y wins against all competitors (strictly: $m_{yj} > 0$ for all $j \neq y$) for some finite $\beta_{\text{eff}} \geq 0$ if and only if*

$$\big(\forall j:\ b_j = 0 \Rightarrow a_j > 0\big) \quad \textit{and}$$
$$\max\Big(0,\ \max_{j:\, b_j > 0} \tfrac{-a_j}{b_j}\Big) \ <\ \min_{j:\, b_j < 0} \tfrac{-a_j}{b_j},$$

where the right-hand side is $+\infty$ when no competitor has $b_j < 0$.

Proof. By Lemma 1 a strict win requires $m_{yj}(\beta_{\text{eff}}) > 0$ for every $j \neq y$, i.e. $a_j + \beta_{\text{eff}} b_j > 0$. This is one-dimensional linear feasibility in $\beta_{\text{eff}} \geq 0$. If $b_j = 0$ the constraint is $a_j > 0$, independent of β_{eff}. If $b_j > 0$ it gives the lower bound $\beta_{\text{eff}} > -a_j/b_j$; if $b_j < 0$ it gives the upper bound $\beta_{\text{eff}} < -a_j/b_j$. The feasible set is the intersection $(L, U) \cap [0, \infty)$ with $L = \max(0, \max_{b_j>0}(-a_j/b_j))$ and $U = \min_{b_j<0}(-a_j/b_j)$, which is nonempty iff $L < U$ and all $b_j = 0$ constraints hold. □

The feasible set is open at L: at $\beta_{\text{eff}} = L > 0$ the binding blocker ties rather than loses. The endpoint $\beta_{\text{eff}} = 0$ is feasible iff y already wins un-edited (all $a_j > 0$), in which case $L = 0$. We write the interval as (L, U) throughout.

Corollary 1 (Unreachability certificate)**.** *If some competitor j has $a_j \leq 0$ and $b_j \leq 0$, then no finite $\beta_{\text{eff}} \geq 0$ makes y win: y is certified unreachable under this value direction and gate. The certificate is computable from h, W, and $\bar{r}$ before any injection.*

Corollary 1 is sufficient, not exhaustive: a target is also unreachable when the interval is empty ($L \geq U$) with no single hard blocker. In the 1B operating envelope (§7), 229 of the 259 certified-unreachable targets at the reference key trip the hard-blocker clause and the remaining 30 are

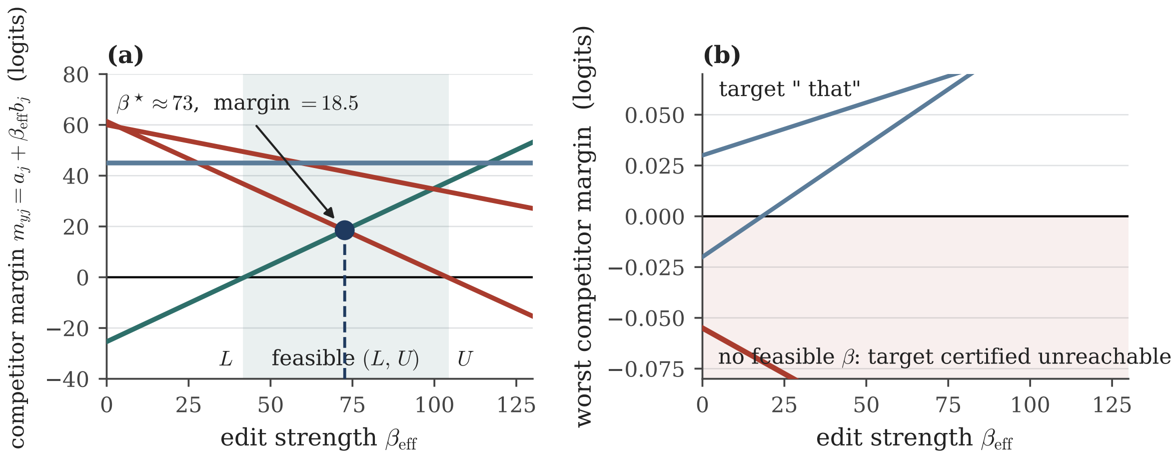

Figure 4: The certificate is one-dimensional linear feasibility (Proposition 1, Corollary 1). Each line is a competitor margin $m_{yj}(\beta_{\text{eff}}) = a_j + \beta_{\text{eff}} b_j$; the target wins where all lie above zero. **(a)** A reachable target: the feasible interval (L, U) is non-empty and $\beta^\star$ maximizes the worst-case margin (values from `betastar.json`). **(b)** An unreachable target (“ that”, blocked by “,”): a competitor with $a_j \leq 0$, $b_j \leq 0$ stays below zero for every β_{eff}, so the certificate refuses it before injection. Representative competitor lines drawn to scale around the measured operating point.

refused by an empty interval. For example the target token “ that” is blocked by the high-frequency competitor “,”: under the deployed value $v = \text{unit}(W_y)$ its worst-case competitor margin is negative (-0.055) with non-positive slope, so no finite $\beta_{\text{eff}} \geq 0$ flips it and the certificate refuses it in advance—while residual synthesis (§7.1) restores a positive margin ($+0.42$) and rescues it (Figure 4).

Observation 1 (Forget). Setting $\text{active}_i = 0$ removes slot i's contribution from $s(\cdot)$ at every future query, so the post-forget behavior equals that of the memory with slot i absent, while a tombstone record persists in the audit log.

Lemma 2 (Off-support fallback, by construction)**.** *At any position where* $\max_i s_i(q) < \tau$*,* (9) *gives* $g(q) = 0$*, hence* $h' = h$ *and* $z' = z$*: the distribution is identical to the frozen trunk.*

This is an immediate consequence of the hard gate rather than a deep result; we state it to make the fallback explicit. Whole-sequence locality on $\mathcal{D}_H$ follows *only if* no held-out query enters the support region $S_\tau = \{q : \max_i s_i(q) \geq \tau\}$. We do not prove this; we *measure* it (zero off-support changes, Section 8.3), and distinguish the per-position by-construction fact (Lemma 2) from the measured whole-sequence locality.

Conjecture 1 (Multi-slot rescue). The realized residual $\bar{r} = r(q)$ is a single query-fixed softmax point, not a free convex combination of slot values. We conjecture that a slot bundle whose fixed mixture realizes a direction with $b_j > 0$ against every blocking competitor can render an otherwise-unreachable target reachable; establishing the achievable residual set is left to future work.

6 Experimental Setup

Protocol. Each scale uses matched-pair pretraining: the baseline and hybrid arms share data, tokenizer, horizon, optimizer, and seed, differing only in the hybrid arm's co-trained memory module. Trunks are trained on FineWeb-derived tokens [39]; the tokenizer is a 65,536-symbol BPE

Canonical axis	Our quantity	Definition
Efficacy / reliability	flip rate	fraction of K facts whose post-edit argmax is the target
Generalization	paraphrase rate	fraction of paraphrases of x that yield the target
Locality (off-support)	holdout changes	argmax changes on held-out text between detached/attached memory
Locality (on-support)	*(pending)*	neighborhood specificity; not yet measured (Section 9)
Reversibility	edit/forget + others-intact	edit flips to a new target; forget reverts; others preserved
Cost	gradient steps	optimization steps required per fact (0 for HLM5)
Auditability	receipt	replay-verifiable, tamper-evident record per operation

Table 1: Crosswalk between the standard editing axes and the quantities measured here. "On-support" neighborhood specificity is a recognized locality test we do not yet run; "holdout changes" measures only off-support locality.

trained on the same distribution, and FineWeb is released under ODC-By 1.0. The 1B run uses width 1792/24L/14H, sequence length 1024, AdamW (betas $0.9, 0.95$, base LR 2×10^{-4}, 2000-step warmup, cosine decay), gradient clip 1.0, weight decay 0.1, and a single fixed seed. Injection studies attach the editable memory to the frozen baseline checkpoint; no gradient steps are taken for any fact. The 17-fact set used throughout the deployed 1B studies is listed in Table 14 (Appendix).

Compute. Pretraining ran on the Leonardo Booster GPU partition (CINECA; 4×A100-64GB nodes) under a EuroHPC AI Factory allocation, in bf16 with periodic validation and checkpointing; each 1B arm trained for 153,000 steps at effective batch 128 × 1024 tokens ($\approx$ 20B tokens per arm; `baseline_train_summary.json`, `hybrid_train_summary.json`). The released 1B checkpoints are the best-validation snapshots within that schedule—step 150,999 (baseline) and step 149,999 (hybrid; `notax_params.json`). The certificate, envelope, and editing studies ran on a single RTX 4000 Ada (20 GB) workstation.

Metrics. We adopt the editing literature's axes and give the correspondence to our measured quantities in Table 1. Off-support locality is measured as argmax preservation on held-out prompts: 8 neutral prompts per fact at 1B (136 prompt–fact checks over 17 facts) and the other facts' key prompts (23 per fact) in the GPT-2 study. All reported numbers are single-seed; sampling uncertainty over facts is reported as bootstrap 95% confidence intervals (2,000 resamples) in Tables 4, 9, and 11, with printed interval endpoints rounded outward to two decimals, while seed-to-seed variance is deferred to E6 (Section 9).

7 From Certificate to Operating Envelope

The operating envelope, not a binary verdict. Section 5 certifies a target as reachable or not; for deployment we report the full feasible interval and a risk label. Throughout this section we work at the exact key, where the measured gate score is 1.00 (so $\beta_{\text{eff}} = \beta$; Figure 7) and the temperature-softmax read concentrates on the injected slot (measured own-slot weight 0.9993 at $T = 0.1$, $K = 17$; `ownslot_weight_certdosed.json`), so slopes are computed directly against the slot value: the certificate is exact in $(\beta_{\text{eff}}, \bar{r})$ and inherits at most this concentration error here.

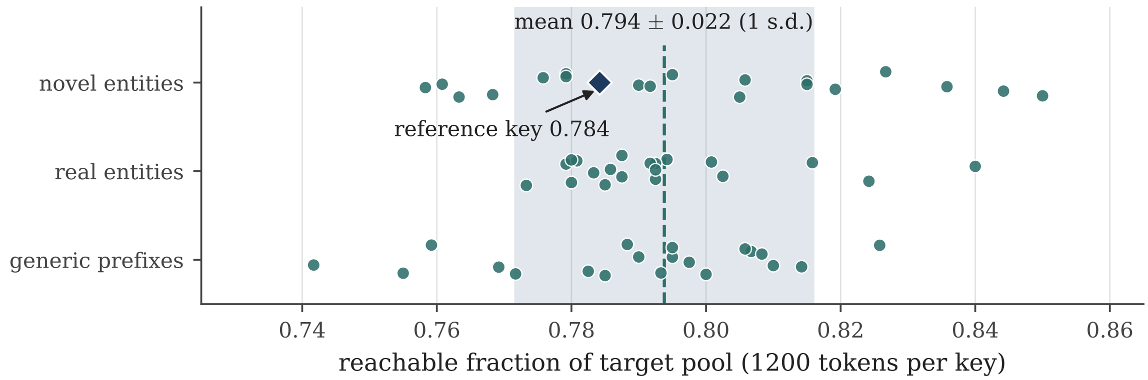

Figure 5: The operating envelope is stable across key contexts. Reachable fraction of the fixed 1,200-token target pool for 60 diverse keys (20 novel-entity, 20 real-entity, 20 generic sentence prefixes): mean 79.4% ± 2.2% (shaded); the reference key of Table 2 (78.4%) is typical. Data: `cert_envelope_multikey.json`.

Risk label	Count	Fraction
Safe	888	74.0%
Narrow	38	3.2%
Brittle	15	1.25%
Unreachable	259	21.6%

Table 2: Operating-envelope certificate over 1,200 word-like single-token targets at a novel-entity key on the frozen 1B trunk, deployed value $v = \mathrm{unit}(W_y)$. Median reachable slack $U - L = 30.2$, computed pre-injection from W and h. Residual synthesis (§7.1) rescues 40/40 sampled unreachable targets.

With the deployed value $v = \mathrm{unit}(W_y)$ and slope $b_j = (W_y - W_j)^\top v$, the feasible edit strengths form (L, U) with $L = \max(0, \max_{b_j>0} -a_j/b_j)$ and $U = \min_{b_j<0} -a_j/b_j$. To each edit we attach the interval (L, U), the chosen β, the slack $U - L$, the binding blocker token, and a label: *safe* (wide slack), *narrow*, *brittle* (small slack), or *unreachable* ($L \geq U$, or a competitor with $a_j \leq 0$, $b_j \leq 0$). Measured on the frozen 1B trunk over 1,200 word-like single-token targets—leading-space single tokens whose text is purely alphabetic with at least three letters; the first 1,200 such token ids—at a novel-entity key (Table 2), 78.4% are reachable (74.0% safe) and 21.6% unreachable, with median reachable slack 30.2. The envelope is computed before any injection, from W and h in one pass over the head matrix—a practical admission test, not only a theorem.

Stability across keys. The reference-key envelope is not an artifact of one hidden state. Repeating the computation with the same 1,200-target pool over 60 diverse key contexts—20 novel-entity templates across several relations, 20 real-entity factual prompts, and 20 generic sentence prefixes—the reachable fraction is 79.4% ± 2.2% (min 74.2%, max 85.0%); a Kruskal–Wallis test finds the three prompt categories statistically indistinguishable ($H = 0.31$, $p = 0.86$; `multikey_category_test.json`). The reference key (78.4%) sits at the 33rd percentile (Figure 5). Per key, a mean of 205 refusals trip the hard-blocker clause of Corollary 1 and 42 have an empty interval.

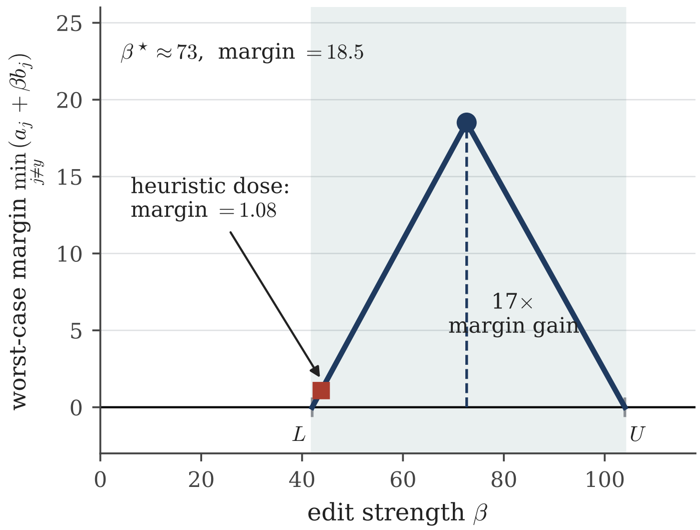

Figure 6: Selecting the deployment strength. The tent is a *representative median envelope* over the 941-fact reachable set, not any single fact's trajectory: the worst-case competitor margin $\min_{j\neq y}(a_j + \beta b_j)$ is concave and piecewise-linear in β, with $\beta^\star$ at its peak. The peak ($\beta^\star \approx 73$, worst-case margin 18.5) and the heuristic margin (1.08 at $\beta = 1.05L+1$, giving the $\approx 17\times$ gain) are *per-fact medians* over that set (`betastar.json`). The feasible-band edges L, U are representative anchors shared with Figure 4, not a measured (L, U) for any one fact.

Choosing the deployment strength. Rather than a heuristic value we select the strength that maximizes the worst-case competitor margin,

$$\beta^\star = \operatorname*{arg\,max}_{\beta \in (L,\, \min(U, \beta_{\max})]} \ \min_{j \neq y} (a_j + \beta b_j),$$

with an explicit cap $\beta_{\max}$ for the case $U = \infty$. The inner objective is concave and piecewise-linear in β; we approximate the maximizer on a dense grid over the capped interval (exact scanning of the breakpoints is equally cheap). On the 1B trunk this lifts the median worst-case margin from 1.08 under the heuristic $\beta = 1.05L + 1$ to 18.5 at $\beta^\star \approx 73$—a $17\times$ improvement in median worst-case margin at identical compute (Figure 6). The receipt records $\beta^\star$, the slack $U - L$, the active blocker $j^\star = \arg\min_j (a_j + \beta^\star b_j)$, and the realized margin $M(\beta^\star)$. Dosing is not cosmetic: on the deployed 17-fact set a global strength of 102.8 exceeds U for four *reachable* facts and silently fails them (efficacy 0.529), while dosing at $\beta^\star$ flips all 13 certificate-reachable facts—efficacy equal to the reachable fraction exactly—at bit-identical off-support locality (§8.6, `hlm5_1b_faithful_certdosed.json`).

A robust certificate under hidden-state drift. The certificate assumes h and the realized read are fixed; in deployment they shift across hardware, precision, batching, or tokenizer revisions. Let $\|\Delta h\| \leq \varepsilon_h$ and let ε_r bound the drift of the *boosted* read $g(q)\,r(q)$ (Remark 1 below derives such an ε_r from Proposition 2's score bound, covering gate-multiplier drift as well as read drift). Every margin is then lower-bounded by

$$m_{yj}(\beta) \ \geq \ a_j + \beta b_j - c_j\,(\varepsilon_h + \beta\,\varepsilon_r), \qquad c_j = \|W_y - W_j\|.$$

Robust reachability is the existence of $\beta \geq 0$ keeping this bound positive for all competitors; equivalently, it is Proposition 1 applied to the shifted coefficients $\tilde{a}_j = a_j - c_j\varepsilon_h$ and $\tilde{b}_j = b_j - c_j\varepsilon_r$. A receipt that is robust-reachable at $(\varepsilon_h, \varepsilon_r)$ stays valid under any serving change within that ball—a portable guarantee rather than a point estimate. The shifted constraints give the robust feasible interval

$$L_{\text{rob}} = \max\Big(0, \max_{j:\, b_j > c_j\varepsilon_r} \tfrac{c_j\varepsilon_h - a_j}{b_j - c_j\varepsilon_r}\Big), \qquad U_{\text{rob}} = \min_{j:\, b_j < c_j\varepsilon_r} \tfrac{c_j\varepsilon_h - a_j}{b_j - c_j\varepsilon_r},$$

where robust reachability is $L_{\text{rob}} < U_{\text{rob}}$ together with $a_j > c_j\varepsilon_h$ for every competitor with $b_j = c_j\varepsilon_r$ (the zero-shifted-slope case), and a competitor with $a_j \le c_j\varepsilon_h$ and $b_j \le c_j\varepsilon_r$ certifies robust impossibility. A margin guarantee is vacuous if the routed slot itself can change under drift, so we add a support-stability certificate.

Proposition 2 (Support stability)**.** *Let $\hat{q} = q/\|q\|$ with $q = (h - \mu)A \neq 0$ and score $s_i(\hat{q}) = \rho_i\,[\max(0, \hat{q}^\top k_i)]^\kappa$, where $\rho_i = \text{active}_i\,\text{relu}(\alpha_i)$, $\rho_{\max} = \max_i \rho_i$, and μ and A are frozen at audit time. If $\|\Delta h\| \le \varepsilon_h$, each score moves by at most $\varepsilon_s := \rho_{\max}\kappa\varepsilon_q$ with $\varepsilon_q := 2\|A\|_{\text{op}}\varepsilon_h/\|q\|$, and the gate keeps the same firing slot whenever $s_{(1)} - \tau > \varepsilon_s$ and $s_{(1)} - s_{(2)} > 2\varepsilon_s$, with $s_{(1)}, s_{(2)}$ the top two scores.*

Proof. Unit normalization is Lipschitz: $\|\Delta\hat{q}\| \le 2\|\Delta q\|/\|q\| \le 2\|A\|_{\text{op}}\varepsilon_h/\|q\| = \varepsilon_q$. The map $u \mapsto [\max(0, u)]^\kappa$ is κ-Lipschitz on $[-1, 1]$ (its derivative is $\kappa u^{\kappa-1} \le \kappa$ on $[0, 1]$ and 0 below), and $|\Delta(\hat{q}^\top k_i)| \le \varepsilon_q$ for unit k_i, so each score moves by at most $\rho_{\max}\kappa\varepsilon_q$. A uniform per-slot drift of ε_s cannot cross the threshold if $s_{(1)} - \tau > \varepsilon_s$, and cannot swap the top two slots if $s_{(1)} - s_{(2)} > 2\varepsilon_s$. ☐

Remark 1 (From score drift to read drift)*.* Proposition 2 bounds the per-slot score drift by $\varepsilon_s = \rho_{\max}\kappa\varepsilon_q$; Lipschitz continuity of the read in the scores turns this into an ε_r. With $p = \text{softmax}(s/T)$ and $r(q) = \sum_i p_i V_i$, the softmax Jacobian $(\text{diag}(p) - pp^\top)/T$ has $\ell_\infty \to \ell_1$ operator norm at most $2/T$, so $\|\Delta p\|_1 \le 2\varepsilon_s/T$ and hence $\|\Delta r\| \le (2\varepsilon_s/T)\max_i\|V_i\|$. For the boosted read $g(q)\,r(q)$, the gate multiplier is itself a score, so it is bounded by $\rho_{\max}$ and drifts by at most ε_s, while $\|r\| \le \max_i\|V_i\|$; whenever the firing slot is unchanged (Proposition 2),

$$\varepsilon_r := \varepsilon_s\Big(1 + \frac{2\rho_{\max}}{T}\Big)\max_i\|V_i\|$$

bounds the drift of the boosted read.

Together the support and margin certificates upgrade the pointwise feasibility theorem into a deployment-time admission test: an edit ships only if its routed slot *and* its winning margin are certified stable within the serving tolerance $(\varepsilon_h, \varepsilon_r)$.

7.1 Residual synthesis rescues unreachable targets

Unreachable cases arise because the value $v = \text{unit}(W_y)$ need not point away from every competitor. Rather than accept this, we solve for a better value without touching the trunk:

$$r^\star = \arg\max_{\|r\| \le 1} \min_{j \neq y} (W_y - W_j)^\top r,$$

a concave program (projected gradient on the soft-min). If its optimum $m^\star > 0$ then every slope is positive and the target is reachable when $r^\star$ is realized as the read—i.e. installed as the dominant slot value, $\bar{r} \approx r^\star$. We verify this at the readout $(z = W(h + \beta r^\star))$ on the 1B trunk: across 40 sampled previously-unreachable targets the naive value $\text{unit}(W_y)$ flips *none*—even at $\beta = 10^5$, the top of the swept β grid (recorded in `run_1b_synth_verify.py`)— whereas the synthesized residual flips *all* 40 at small strength (median $\beta = 20$, maximum 50). Off-key locality is preserved because the whitened gate fires only at the key (gate selectivity, below) independent of the stored value;

on-support behavior of the synthesized value at the key—and the full memory-path replication of this rescue—is deferred to E3 (§9). The rare-token failures are therefore failures of the *value choice*, not of head geometry, repairable by a small external optimization over W and h with no gradient through the trunk. This is complementary to Conjecture 1, which concerns fixed mixtures of *existing* slots; synthesizing a fresh value sidesteps, rather than settles, that question.

Certified refusal. When the envelope is empty the system should not fail silently: it emits a *refusal receipt*—the unreachable target, the blocking competitor token, its (a_j, b_j), the empty feasible interval, and (when synthesis succeeds) the proposed alternative residual. Refusal becomes a first-class, audited outcome: the 259 certified-unreachable targets in the envelope (Table 2) are exactly such receipts—229 blocked by a hard competitor (“,” is the most frequent blocker in the sampled cases), 30 by an empty interval—all flagged *before* injection, with a synthesized rescuing residual proposed for all 40 sampled cases.

Gate selectivity (collision ROC). Off-support exactness is automatic when the gate does not fire; the real deployment risk is accidental firing on boundary text. After injecting capital facts we measure the maximum gate score across probe categories. The whitened degree-κ gate is perfectly separable (Figure 7): exact keys score 1.00 while paraphrases, same-subject/other-relation prompts, single-character typos, and random held-out text all score 0.00, so any threshold $\tau \in (0, 1)$ yields zero collisions with full separation between the score populations. This makes the whole-sequence locality of §5 a *measured* property rather than only a hard-gate tautology. In plain terms: at this operating point the whitened gate is functionally an exact-key cache, and the interesting regime for generalization is the relaxed gate of §8.6. The flip side is brittleness: the same selectivity means a typo or paraphrase of the key also scores 0.00, consistent with the exact-key scope (§8.5). A finer-grained sequential-edit stress test (create/edit/forget/re-add up to 1,024 slots) confirms flat per-operation latency ($\approx$ 2.3 ms/query, $\approx$ 0.2 ms/inject at steady state) and exposes a softmax read-dilution effect at large slot counts. Its functional outcomes are a *negative* result for the auto-boost configuration it used: under globally boosted synthetic-entity keys the create flip rate is only 0.115–0.25, others-intact is only 0.117–0.25, edit and re-add succeed on 0.0 of operations at every K, and tombstone verification degrades from 1.0 at $K \leq 256$ to 0.667 at $K = 512$ and 0.451 at $K = 1{,}024$ (`seq_edit_stream.json`). This harness predates certificate dosing (a single global boost rather than per-fact $\beta^\star$), so we read it as evidence against *undosed* sequential editing rather than as the mechanism’s ceiling; the redesigned, certificate-dosed sequential study is E5 (§9).

Beyond argmax. A binary flip understates deployment behavior. On a separate 11-fact capital/science probe set—evaluated with the raw additive read and per-fact auto-calibrated β, without the deployed gate, so as to isolate the read (flip rate 0.818, 9/11)—the deployed value wins the argmax with a median post-edit logit margin of only 1.51 and target probability mass $\approx$ 0.50 (Table 3): the edit is correct but not dominant, so it can be brittle under sampling, multi-token continuation, or downstream reformatting. (The deployed pipeline’s efficacy—0.765 certificate-dosed, 0.529 under the global-boost ablation (§8.6)—is measured on a different fact set with the whitened gate; the numbers are not in tension.) Deployed receipts should therefore record the post-edit margin, rank, and probability, not only the flip.

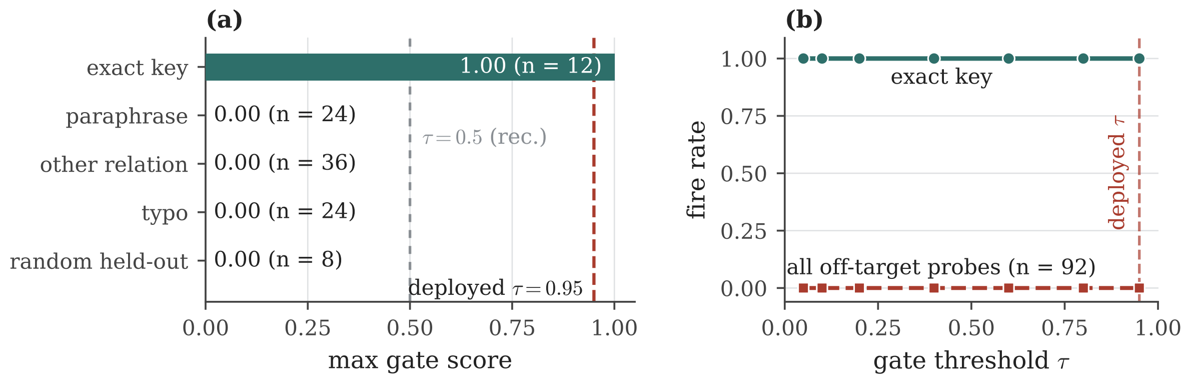

Figure 7: Gate selectivity on the frozen 1B trunk. **(a)** The whitened degree-κ gate scores exact keys at 1.00 and every off-target probe category—paraphrases, other-relation prompts, single-character typos, random held-out text—at 0.00. **(b)** The exact-key and off-target fire rates stay fully separated across the whole threshold range: separation is perfect, so any $\tau \in (0, 1)$—including the deployed $\tau = 0.95$ (dashed)—yields zero collisions. Data: `gate_roc.json`.

8 Results and Analysis

Table 4 summarizes HLM5 against the canonical axes; the controlled comparisons against logit-bias, fine-tuning, ROME, and retrieval baselines follow in Section 8.6 (Tables 9 and 11). A head-to-head comparison against MEMIT, MEND, SERAC, and GRACE on CounterFact/zsRE has not yet been run; it is the first item of the falsification plan (E1, Section 9).

8.1 The co-trained memory imposes no language-model tax

The perplexity cost of co-training the memory module is below 0.12% at both scales and is no larger at 1B than at 136M (Table 5, Figure 8): +0.018% at 136M and −0.116% at 1B. Both deltas are single-seed and within an unmeasured run-to-run noise band; we claim only that any tax is below 0.12%, not an improvement. The head-reachable fraction of the vocabulary grows from 1.85% at 136M (`g2c_rare.json`) to 10.27% at 1B, consistent with a larger, better-conditioned head exposing more flippable directions.

8.2 Relation to energy-based memory programs

The mechanism of Section 4 is the editing fragment of a broader program that treats a network's energy landscape as something to be *programmed* rather than retrained—"energy minima as a programming language" [41]: slots are attractor basins, and operations *survey*, *inject*, *strengthen*, *forget*, *capture*, *blend*, and *transplant* them [40]. Table 6 gives the correspondence, and the piece the operator set otherwise lacks is the certificate: each *inject* is accepted or refused in advance by Proposition 1. Prior work realizes the energy-landscape view *inside* the network dynamics [24, 42, 18]; here an external, additively coupled associative memory suffices on a frozen softmax transformer. Porting the certified operator set to a natively energy-based language model (HLM3) is future work (E8, §9).

Edit (target)	flip	β	logit margin	target prob	rank
France → London	✓	20	1.26	0.77	0
Italy → Paris	✓	20	3.82	0.82	0
Spain → Rome	✓	20	3.11	0.50	0
Diamonds → iron	✓	5	1.38	0.26	0
Sun rises → west	✓	5	0.04	0.47	0
Japan → Berlin	×	—	−341.3	0.00	9089
Germany → Tokyo	×	—	−491.1	0.00	13304

Table 3: Beyond argmax: per-edit strength on the frozen 1B trunk (representative rows from `headgeom_metrics.json`; 11 facts, flip rate 0.818). Successful flips win with a *median* logit margin of only 1.51 at target probability ≈ 0.50—correct but not dominant—while head-geometry failures sit thousands of ranks from the target. Deployed receipts should record margin, probability, and rank, not only the flip. Facts are shown as subject → counterfactual target; rank is 0-indexed (0 = argmax).

Method (setting)	Efficacy	Generalization	Locality (off-supp.)	Reversibility	Grad. steps
HLM5 (1B, deployed τ=0.95, certificate-dosed, 17 facts)	0.765 (13/17) [0.52, 0.95]	0.020 [0.00, 0.06]	0 changes (1.000 [1.00, 1.00])	forget exact‡	0
HLM5 (1B, deployed, + synthesis rescue)	1.000 (17/17) [1.00, 1.00]	0.020 [0.00, 0.06]	0 changes (1.000 [1.00, 1.00])	forget exact‡	0
HLM5 (1B, loose gate $\tau_{\cos}$=0.40, 17 facts)	0.765 (13/17) [0.52, 0.95]	0.353 [0.19, 0.53]	0.955 [0.92, 0.99]	forget exact‡	0
HLM5 (GPT-2, controlled, 24 facts)	1.000 (24/24) [1.00, 1.00]	0.556 [0.40, 0.71]	0.949 [0.92, 0.97]	—	0

Table 4: Summary of HLM5 operating points, single seed. Efficacy is flip rate; Generalization is paraphrase rate; off-support Locality is held-out argmax preservation (Section 6). Brackets: bootstrap 95% CIs (2,000 resamples). The loose replication gate $\tau_{\cos}$=0.40 thresholds the centered cosine directly, before the degree-κ power and without whitening. ‡Forget is exact by construction (Obs. 1); the tombstone is verified at single-fact scale (`read_and_memorize_receipts.json`, `edit_receipts.json`), while tombstone verification at scale degrades in the undosed sequential stress test (`seq_edit_stream.json`, §7) and, with sequential edit/re-add efficacy, is deferred to E5 (§9). Certificate-dosed = each slot stored at its certified $\beta^\star \in (L, U)$; the global-boost dosing ablation (0.529) is in Table 11. Standard-benchmark baselines are pending (E1, §9).

Running the operator set directly on the co-trained memory of the frozen hybrid 1B checkpoint—reusing the same module, no retraining—gives the measurements of Table 7. Two caveats keep these results in proportion. The inject/strengthen/forget rows verify implementation invariants of the score function (7)—they hold by construction—so the informative measurements are the survey statistics of the trained store and the off-target guard. The store holds 256 attractor basins at unit self-energy with maximum pairwise coherence 0.10 (mean 0.019) and maximum crosstalk $\sim 10^{-5}$. A *capture+blend* of two concepts retrieves a value at cosine 0.89 to each source (a concept-pair-dependent constant: the sources here sit at cosine 0.59 to each other). For the guard, the injected slot's maximum gate score on held-out probes is 2.7×10^{-2} (mean 2.9×10^{-3})—35× below the firing threshold $\tau = 0.95$, so the gate never fires off-target; the store's pre-edit background gate energy on the same probes is $\sim 3.5 \times 10^{-6}$. Reproduced by `run_hybrid_energy_ops.py` (→ `hybrid_energy_ops.json`).

8.3 On-trunk injection: envelope, deployment, and exact fallback

Attaching the editable memory to the frozen baseline trunk realizes the operating envelope of Section 7: the certificate labels each target in advance, the split is stable across keys, and the readout-level synthesis rescue covers the sampled refusals; the deployed efficacy and dosing numbers

Scale	Params (base → hybrid)	Val. PPL base	Val. PPL hybrid	Δ
136M	136.17M → 137.75M	40.158	40.166	+0.018%
1B	1044.68M → 1052.03M	18.013	17.992	−0.116%

Table 5: Matched-pair no-tax study (single seed). The co-trained memory adds ≈ 7.3M parameters at 1B. Parameter counts at both scales are counted from the local checkpoints (`notax_params.json`); the 1B baseline PPL (18.013) is reproducible from the released checkpoint (`hlm5_1b_table_row.json`), and the remaining PPLs are recorded from the matched-pair training runs' gate reports (`g2a_no_tax.json` at 136M, `g3a_no_tax.json` at 1B). The 1B PPLs are best-validation snapshots within the shared 153,000-step schedule, taken at step 150,999 (baseline) and step 149,999 (hybrid) (`baseline_train_summary.json`, `hybrid_train_summary.json`, `notax_params.json`).

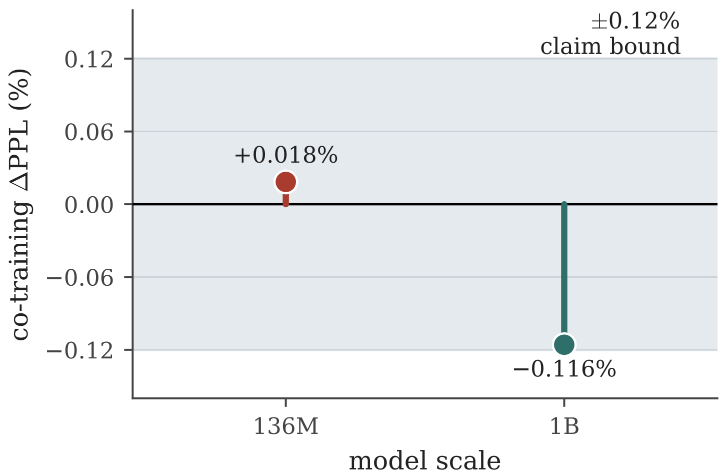

Figure 8: Signed co-training perplexity delta, (hybrid − baseline)/baseline, per scale: +0.018% at 136M (co-training marginally worse) and −0.116% at 1B (co-training better). The shaded band is the paper's ±0.12% no-tax claim bound; both scales fall inside it. Data: `g2a_no_tax.json`, `g3a_no_tax.json`, `notax_params.json`.

are the controlled comparison of Table 11 (§8.6). What deployment adds is the *exact fallback*: under the conservative gate $\tau = 0.95$ the gate fires on none of the held-out or paraphrase prompts (8 held-out prompts per fact, §6), so $h' = h$ off-key (Lemma 2) and the output is *bit-identical* to the frozen trunk—off-support locality 1.000 is exact, not approximate. At a loose replication gate ($\tau_{\cos} = 0.40$, globally dosed) efficacy is 0.765 while measured locality is 0.955 (Table 4). Reversibility is handled outside the trunk: forget is exact by construction (Observation 1) with a tombstone verified at single-fact scale (`read_and_memorize_receipts.json`), and no per-fact gradient is taken. Efficacy is bounded by the reachability frontier (Corollary 1), not by collateral damage—the central operating characteristic of the certified injection.

8.4 Mechanism capacity (synthetic)

To characterize the memory module independent of any trunk, we study two synthetic settings. *These are not 1B-trunk results.* In a routed key–value memory with no language model, the number

Energy-language operator	HLM5 realization	Certified by
inject / strengthen	add slot (k, v, α); set β	Prop. 1 (β-feasibility)
forget / remove	$\text{active}_i \leftarrow 0$, tombstone	Obs. 1 (exact removal)
survey / measure energy	gate score $\max_i s_i(q)$	Lemma 2 (off-support identity)
capture / blend / transplant	build v from W, h; copy slots	value synthesis (§7.1); operational checks only
guard (safety)	hard support gate τ; receipt	Prop. 2 (support stability)

Table 6: HLM5 as the *certified* fragment of the energy-language operator set. The first two rows and the guard carry a-priori guarantees; capture/blend/transplant are validated only operationally (Table 7).

Operator	Measured effect on the hybrid memory (gate energy g)
survey	256 basins, max coherence 0.10 (mean 0.019), crosstalk $\sim 10^{-5}$, self-energy 1.00
inject	captured key: g $0.00 \to 1.00$
strengthen/weaken	$\alpha \times 0.5$: g $1.00 \to 0.50 \to 1.00$
forget	tombstone: g $1.00 \to 0.00$
capture+blend	50/50 blend retrieves at $\cos 0.89$ to each source
transplant	basins copied to a fresh store keep $g = 1.00$
guard (locality)	post-edit off-target gate score $\leq 2.7 \times 10^{-2}$ ($35\times$ below τ; zero off-target firings)

Table 7: Energy-language operators run on the co-trained memory of the frozen hybrid 1B checkpoint. Inject/strengthen/forget verify score-function invariants (by construction); survey, blend, and the guard are the substantive measurements (`hybrid_energy_ops.json`).

of distinct pairs storable at ≥ 0.99 retrieval accuracy is bounded by crosstalk, not accuracy: capacity is $(0.74–0.82)\,M$ pairs—i.e., 74–82% of the M=128 slots, across two independent searches (95/128 and 105/128)—when the dimension is small (D=64), and saturates at the slot count M once $D \geq M$ (128 at M=128, D=128; 256 at M=256, D=256). Artifacts: `hlm5_capacity_search_report.json` (per-point sweeps under `results/capacity_search/`) and `hlm5_scale_suite_report.json`. In a from-scratch toy LM over a synthetic context-to-token map (Table 8), injection is clean up to a collateral wall: at D=64 others-intact collapses to 0.000 at K=96 while audit stays 1.000; at D=256 the default strength under-applies (efficacy $\leq$ 0.969 without corrupting others), and raising β restores perfect efficacy through K=80. The reading is that collateral capacity scales with dimension and edit strength must be matched to dimension—behavior the on-trunk ablation (deferred, Section 9) should confirm.

8.5 Generalization and negative results

Edits are exact-key, not entity-level. Under the deployed 1B gate the edit does not transfer to paraphrases (generalization 0.020, §8.6); on pretrained GPT-2 even the deliberately loosened controlled gate lifts paraphrase generalization only to 0.556 over 24 facts at efficacy 1.000 (Table 9). The additive memory is also not an in-context recall mechanism: on a synthetic multi-query associative-recall probe, attaching it degrades a plain transformer from 0.816 to 0.118 recall at 64 key–value pairs (both are perfect at 32; `incontext_mqar.json`). We report these so that the operating point is unambiguous: HLM5 governs exact single-token edits and does not deliver paraphrase-robust or recall-style memory. Weak paraphrase transfer of a fixed additive intervention is consistent with reliability critiques of steering-vector edits [44], and we make no claim about logically entailed facts,

K	D=64 (flip/audit/intact)	D=256, β=25 (flip)	D=256, β=70 (flip/audit/intact)
16	1.000/1.000/1.000	0.938	1.000/1.000/1.000
32	1.000/1.000/1.000	0.969	1.000/1.000/1.000
48	0.979/1.000/1.000	0.896	1.000/1.000/1.000
64	0.984/1.000/1.000	0.891	1.000/1.000/1.000
80	0.988/1.000/1.000	0.800	1.000/1.000/1.000
96	0.979/1.000/**0.000**	—	—

Table 8: Synthetic injection-vs-K on a toy LM (single `TransformerEncoderLayer`), *not* the 1B trunk. At D=64 a collateral wall appears at K=96 (others-intact 0.000); at D=256 adequate edit strength is required for efficacy. Data: `lm_kb_inject.json` (D=64), `lm_kb_inject_d256.json` (D=256, β=25), `lm_kb_inject_d256_b70.json` (D=256, β=70); β=25 is the producing script's default, recorded in the script, not the JSON.

the ripple-effect axis of Cohen et al. [3].

8.6 Controlled comparison against baselines

To probe the most direct objection—whether the additive gated read differs from a key-conditioned logit bias—we run a controlled study on pretrained GPT-2 (124M) over 24 single-token counterfactual facts, comparing a simplified single-slot HLM5 read against four baselines under identical facts, gate, and metrics (Table 9): a static logit-bias applied under the *same* hard gate, full fine-tuning (FT-full), constrained last-block fine-tuning (FT-L), and a retrieval control (the oracle fact prepended in context). At equal efficacy (1.000) and near-equal locality, HLM5 generalizes to paraphrases significantly better than the matched logit-bias (0.556 vs 0.347; McNemar paired test over paraphrase trials, 15 vs 0 discordant, $p = 6 \times 10^{-5}$): on GPT-2, the additive read carries information a target-logit bump does not. Fine-tuning reaches higher generalization only by collapsing locality (FT-full 0.118, FT-L 0.752 vs HLM5 0.949), reproducing the specificity cost of weight editing. ROME, the canonical locate-and-edit baseline (run via EasyEdit), generalizes strongly—0.972 at locality 0.771 on the capital subset—but it mutates trunk weights and optimizes a value vector by gradient, whereas HLM5 is zero-gradient and preserves locality far more tightly (0.949). An independent re-implementation ($n = 18$ facts) reproduces ROME's profile (generalization 0.815, locality 0.854). Plotting every method in the generalization–locality plane makes the trade-off geometric: no method reaches the upper-right corner where both are high (Figure 9).

The GPT-2 comparison deliberately loosens the gate so that it fires on paraphrases. We next ask what the *deployed* pipeline does on the frozen 1B trunk—whitening, the real degree-κ read, and the conservative gate $\tau = 0.95$ (Table 11). At this operating point the gate fires on *none* of the paraphrase prompts (fire rate 0.00) in every arm: HLM5 is strictly exact-key, with paraphrase generalization 0.020—identical to the matched logit-bias (McNemar $p = 1.0$)—and *perfect* locality (1.000, bit-identical held-out logits) for all methods. Efficacy is where dosing matters (Figure 10). A global edit strength (auto-boost 102.8, calibrated by the harness's pre-injection probe as 1.5× the smallest strength that flips every flippable fact under a loose, positive-slope-only test, plus 1) reaches only 0.529: it overshoots the certified upper bound U for four *reachable* facts (e.g. " Rome" at $U = 63.9$, " Saturn" at $U = 39.2$) and silently fails them—more strength is not more efficacy; the interval is real. Dosing each edit at its certified $\beta^\star$ flips exactly the certificate-reachable set (0.765, 13/17, with 4/4 certified-unreachable facts failing as predicted—zero misclassification), and residual synthesis (§7.1) rescues those four (1.000, 17/17). Two conclusions follow. First,

Method (GPT-2, 24 facts)	Efficacy	Generaliz.	Locality
HLM5 (additive gated read)	1.000 [1.00, 1.00]	0.556 [0.40, 0.71]	0.949 [0.92, 0.97]
Static logit-bias (same gate)	1.000 [1.00, 1.00]	0.347 [0.22, 0.49]	0.980 [0.96, 1.00]
ROME (EasyEdit)†	1.000	0.972	0.771
FT-full	1.000 [1.00, 1.00]	0.889 [0.80, 0.96]	0.118 [0.08, 0.17]
FT-L (constrained)	1.000 [1.00, 1.00]	0.847 [0.76, 0.94]	0.752 [0.70, 0.81]
Retrieval (control)	0.667 [0.45, 0.84]	0.375 [0.23, 0.53]	0.531 [0.49, 0.57]

Table 9: Controlled comparison on GPT-2 (124M), single seed. Brackets: bootstrap 95% CIs (2,000 resamples; `gpt2_table1_v2.json`). HLM5 vs static logit-bias on paraphrase generalization: McNemar 15/0 discordant, $p = 6 \times 10^{-5}$. †ROME (run via EasyEdit) requires the subject in the prompt, so it is evaluated on a 12-fact capital subset (other rows: the 24-fact mixed set) and its artifacts record point estimates only (no CIs); an independent re-implementation ($n = 18$ facts) reproduces the profile (generalization 0.815, locality 0.854).

the additive read's generalization edge over a logit patch appears on GPT-2 but *not* on the 1B trunk: paraphrase generalization stays read-limited (below 0.3) across a sweep of the gate threshold (measured under global-boost dosing), so the edge is model-specific, not a general property of the read. Second, the properly dosed pipeline *matches* the gate-matched logit bias on every axis measured here (efficacy 1.000, generalization 0.020, locality 1.000): the mechanism's remaining value over the simplest certifiable variant is the a-priori admission test, the certified dose, the synthesis rescue, and the audit trail—and, against fine-tuning, the substantive trade stands: fine-tuning generalizes (Table 9) but collapses locality and requires gradients. Loosening the gate to buy generalization—and absorbing the locality cost it brings—is future work, alongside the remaining EasyEdit MEMIT/MEND/SERAC baselines (Section 9).

8.7 The certificate predicts editor behavior on CounterFact

Because the certificate is computed from (W, h) alone, it is a property of the *model*, not of our memory—so it can be tested against editors that do not use it. With success criteria fixed in advance, we computed certificate features for the first 1,000 CounterFact records [33] on GPT-2-XL and ran ROME, GRACE, and constrained fine-tuning (via EasyEdit [46]) on the first 300, single-edit-from-fresh-model, using each record's native paraphrase and neighborhood prompt sets (Table 10). The correlation analysis is exploratory: it spans 3 editors × 3 certificate features × 2 outcomes (18 Spearman tests, all reported in Appendix C with Benjamini–Hochberg FDR control at $q = 0.05$); the primary pre-specified predictor is the required strength L, and the two margin features are reported for completeness. Three findings. **(i) The reachability frontier is model-dependent.** On GPT-2-XL all 1,000 targets are certified reachable—a measured 99.82% of its tokens (50,166/50,257, fp32 self-NN under the tied head; `headgeom_gpt2xl.json`) are their own head nearest-neighbor—in sharp contrast to the 1B trunk's 21.6% refusal rate (§7); consequently every editor succeeds almost everywhere (efficacy 0.97–1.00) and rewrite *failure* is unpredictable from the certificate (a ceiling effect, and the registered failure-prediction criterion is not met). **(ii) The certificate predicts generalization and collateral before the edit.** The required strength L predicts paraphrase transfer for ROME ($\rho = -0.16$, $p = 0.005$) and, directionally, for FT ($\rho = -0.13$, $p = 0.023$); FT's strongest paraphrase predictor is instead the post-heuristic dosed margin (`margin_after_heur`: $\rho = -0.16$, $p = 0.007$). GRACE also shows a nominally significant L correlation ($\rho = -0.16$, $p = 0.007$), but we do not count it as a paraphrase-prediction result: GRACE's aggregate paraphrase success is 0.005 (Table 10), a degenerate base rate over which a rank

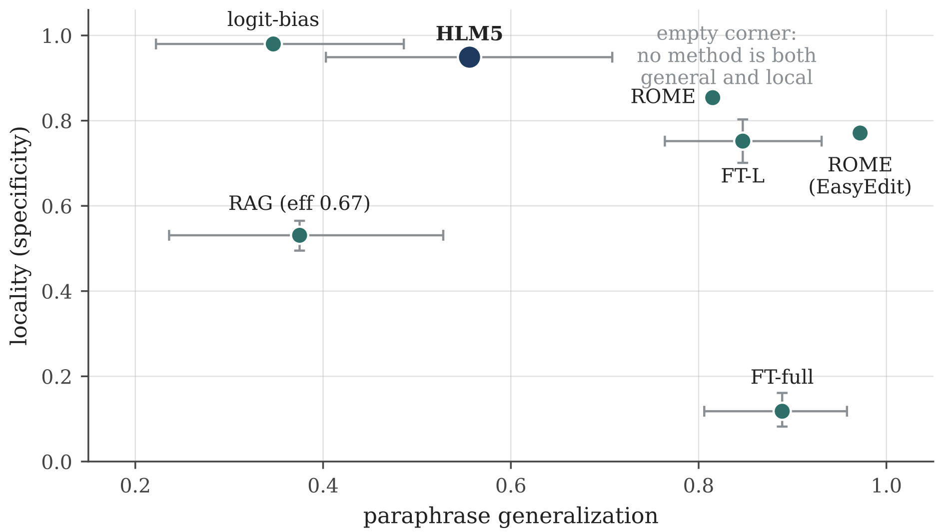

Figure 9: The edit-quality frontier. Every method from Table 9 placed in the (paraphrase generalization, locality) plane. Whiskers are 95% bootstrap CIs on both axes for the five `gpt2_table1_v2.json` methods; the two ROME points record means only (no CIs). No method occupies the upper-right corner: high paraphrase generalization and high locality are not achieved together—each method trades one for the other. Data: `gpt2_table1_v2.json`, `rome_gpt2.json`, `rome_easyedit_gpt2.json`.

correlation carries little evidence. ROME's paraphrase success drops from 0.82 on the lowest-L quartile to 0.62 on the highest (Figure 11). The worst-case certificate margin (`margin_at_beta_star`) predicts neighborhood damage for ROME ($\rho = -0.17$, $p = 0.0035$) and GRACE ($\rho = -0.18$, $p = 0.0014$). All six correlations quoted in this paragraph survive Benjamini–Hochberg FDR control at $q = 0.05$ over the full 18-test grid (10 of 18 cells survive; Appendix C). Across editors, high-L targets generalize worse but bleed less—the certificate measures how embedded a target is in the model's output geometry, and that embedding governs the generalization/collateral trade regardless of editing method. Effect sizes are modest ($|\rho| = 0.13$–0.23) on one model; scaling this study is future work. **(iii) The exact-key trade is a family signature.** GRACE—the closest published wrapper editor—scores efficacy 1.000, paraphrase 0.005, and the best neighborhood preservation (0.776) over the 300 records: the same profile as our deployed gate (Table 11). Gate-protected editors buy specificity with exact-key scope; this is a property of the design point, not a defect of one system.

8.8 Receipts

Each operation emits a replay-verifiable receipt recording the operation, the key prompt and hidden-state hash, the target, the memory hyperparameters, the reachability certificate, and six tamper-evident hashes (query, key, value, registry, trunk, proof). In a small-scale harness the receipt verifies on replay and the harness detects value, prediction, and prompt tampering (`edit_receipts.json`); the corresponding edit applies with zero gradient steps and no collateral. This validates the audit machinery at small scale; scaling it to the trunk's edit stream is an engineering task rather than a new method.

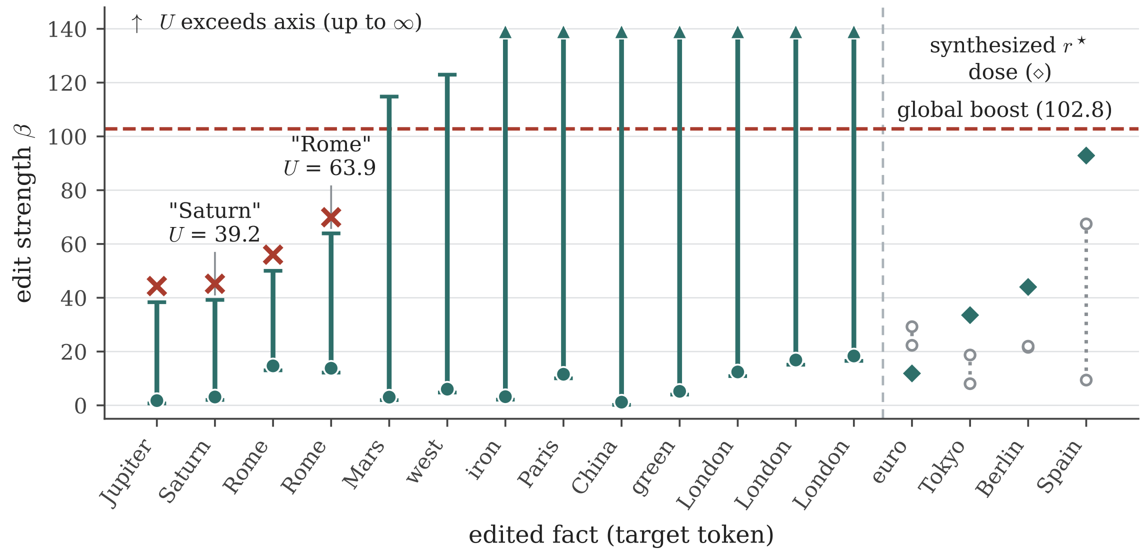

Figure 10: The certificate doses the edit. Certified intervals (L, U) with the deployed dose $\beta^\star$ (dots) for the 13 certificate-reachable facts (sorted by U; arrows: U beyond the axis), and synthesis doses (diamonds) for the four naive-unreachable facts (right of the separator). The global boost (102.8, dashed) exceeds U for four reachable facts (red $\times$)—exactly the silent failures behind efficacy 0.529; dosing inside the interval flips 13/13, and synthesis rescues 4/4. Grey dotted segments show the inverted naive window ($L > U$) that certifies these targets unreachable under unit(W_y). Data: `hlm5_1b_faithful_certdosed.json`.

9 Limitations and Falsification Plan

We frame the open gaps as named experiments with explicit pass criteria, so the claims are falsifiable rather than merely incomplete.

- **E1: standard-benchmark baselines.** Run MEMIT, MEND, SERAC, PMET [30], and AlphaEdit [10] against HLM5 on CounterFact and zsRE at scale ($\geq$10k records, multi-seed). *Pass:* HLM5's locality and zero-gradient cost stay competitive while its certificate predicts each method's behavior; *fail:* a baseline dominates HLM5 on locality at equal cost. (Partially closed: reference ROME on GPT-2, §8.6; ROME/GRACE/FT single-edit on 300 CounterFact records with neighborhood specificity, §8.7.)
- **E2: on-support neighborhood specificity.** Beyond the perfectly separable gate ROC (§7), evaluate CounterFact neighborhood prompts (same subject/other relation; same relation/other subject) for HLM5 itself; the baseline editors' neighborhood scores are now measured (§8.7: ROME 0.639, GRACE 0.776, FT 0.680). *Pass:* neighborhood success $\geq$ 0.95 with the gate firing only on the key.
- **E3: residual-synthesis locality and cost.** Synthesis rescues 40/40 unreachable targets at the key (§7.1); test whether $r^\star$ preserves off-target locality with bounded β. *Pass:* rescued edits keep zero off-support changes at finite, calibrated β.
- **E4: generalization and multi-token targets.** The deployed gate fires on 0% of paraphrases (§8.5). Test a relaxed but certified gate and multi-token targets (cf. sequential decom-

Editor (GPT-2-XL, 300 records)	Efficacy	Paraphrase	Neighborhood	Rewrite fails
ROME	0.987	0.737	0.639	4
GRACE	1.000	0.005	0.776	0
FT (constrained)	0.967	0.068	0.680	10

Table 10: Standard editors on the first 300 CounterFact records (single edit from a fresh model; argmax metrics; neighborhood = fraction of neighborhood prompts where the true target still wins). GRACE reproduces the exact-key profile of the deployed HLM5 gate. Certificate features computed pre-edit predict paraphrase transfer (ROME, $p < 0.01$; FT directionally, $p = 0.02$) and neighborhood damage (ROME, GRACE; $p < 0.01$); the full 18-cell correlation grid with FDR control is in Appendix C; artifacts: `cert_counterfact_gpt2xl.jsonl`, `editors_counterfact_gpt2xl.jsonl`, `cert_vs_editors_analysis.json`.

Method (1B, deployed gate τ=0.95)	Efficacy	Generaliz.	Locality
HLM5, certificate-dosed ($\beta^\star$)	0.765 (13/17) [0.52, 0.95]	0.020 [0.00, 0.06]	1.000 [1.00, 1.00]
HLM5, certificate-dosed + synthesis	1.000 (17/17) [1.00, 1.00]	0.020 [0.00, 0.06]	1.000 [1.00, 1.00]
HLM5, global boost (dosing ablation)	0.529 (9/17) [0.29, 0.77]	0.020 [0.00, 0.06]	1.000 [1.00, 1.00]
Static logit-bias (same gate)	1.000 [1.00, 1.00]	0.020 [0.00, 0.06]	1.000 [1.00, 1.00]
Retrieval (control)	0.294 [0.11, 0.53]	0.176 [0.05, 0.34]	0.507 [0.42, 0.59]

Table 11: Deployed pipeline on the frozen 1B trunk (whitening, degree-κ read, gate $\tau = 0.95$; `hlm5_1b_faithful_certdosed.json`; logit-bias and retrieval rows: `hlm5_1b_faithful.json`). Brackets: bootstrap 95% CIs (2,000 resamples). The gate fires on 0% of paraphrases in every arm, so HLM5 is exact-key at perfect locality (bit-identical held-out logits). Dosing each edit at its certified $\beta^\star$ flips exactly the certificate-reachable set (13/17); residual synthesis rescues the four certified-unreachable facts (17/17); the global boost (102.8) is the ablation—it overshoots the certified upper bound U for four reachable facts and silently fails them.

position of long-form knowledge [22]). *Pass:* paraphrase efficacy rises with bounded locality loss; otherwise the exact-key scope is confirmed by design.

- **E5: sequential-edit efficacy at scale.** Latency is flat to 1,024 edits (§7); a clean reachable-target stream must measure create/edit/forget/re-add efficacy, interference, and read dilution, against the sequential-editing degradation documented for weight editors [14] and the lifelong-editing baselines [15, 47, 13]. *Pass:* others-intact ≥ 0.99 with exact tombstone and rollback.
- **E6: stronger metrics and seeds.** Replace argmax-flip with logit margin, rank, probability mass, KL shift, and exact-answer and long-form generation [43] over ≥ 5 seeds with confidence intervals. *Pass:* the reachability and locality results hold under these metrics (median post-edit margin is currently only 1.51 at probability ≈ 0.50, §7).
- **E7: scale.** Repeat the gates, operating envelope, and certificate at 3B. *Pass:* the certificate's predictive accuracy and the no-tax result hold at 3B.
- **E8: native energy-based trunks.** Port the certified operator set (Table 6) from the additive external memory to a natively energy-based, polynomial-Hopfield language model (HLM3, the predecessor Hopfield-energy architecture; cf. energy-based transformer blocks [18]). *Pass:* the reachability certificate and gate selectivity transfer at comparable locality.

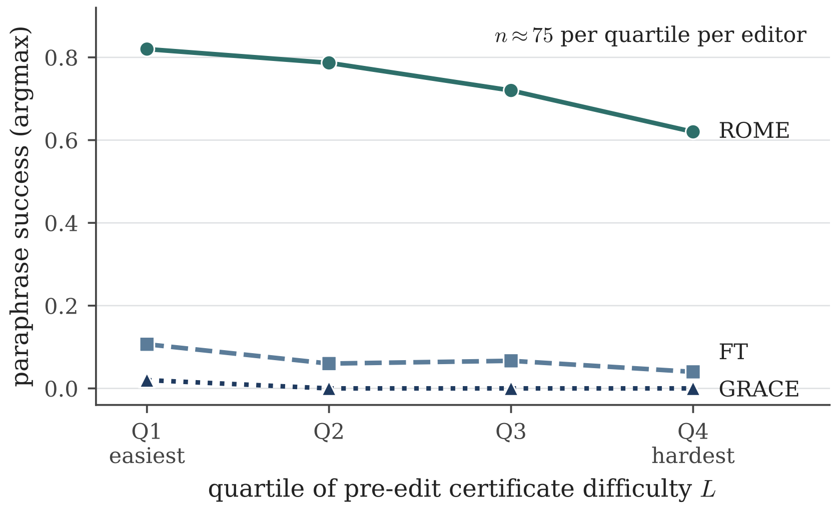

Figure 11: Pre-edit certificate difficulty predicts post-edit generalization for editors that do not use the certificate. Mean paraphrase success by quartile of the certificate's required strength L ($n \approx 75$ records per quartile per editor, GPT-2-XL, 300 CounterFact records): ROME falls from 0.82 to 0.62; FT declines at a lower level; GRACE is exact-key throughout.

10 Conclusion

HLM5 is not a general-purpose knowledge editor; it is a certified, auditable, exact-key edit layer for frozen language models. Its central contribution is that edits can be accepted or refused *before* injection: a closed-form reachability certificate labels single-token edits reachable or unreachable, with locality controlled by a hard support gate and reversibility handled outside the trunk. The constraint of an additive pre-head write buys this certificate—an operating envelope with risk labels that is stable across diverse key contexts—together with an exact off-support fallback by construction and a co-trained memory layer whose perplexity cost is below measurement noise; residual synthesis further shows that the rare failures are failures of value choice, not head geometry, and certificate-governed dosing shows that apparent efficacy failures can be dosing failures: dosed at $\beta^\star$ with synthesis rescue, the deployed pipeline reaches full efficacy at bit-identical off-support locality. Against a static logit patch the boundary is then not editing power—the two match on every axis measured here—but contract: HLM5 is preferable when the a-priori certificate, the certified dose, residual-space rescue, an edit/forget registry, and an audit trail matter. The scope is the single-token counterfactual fact; the open gap—head-to-head editing baselines on standard benchmarks—is the first item of the falsification plan (E1).

Ethics and Broader Impact

Audit integrity is not semantic truth. A receipt proves that an operation occurred and can be replayed; it does not prove the stored fact is true, current, authorized, or mutually consistent. A deployment therefore needs a source-of-truth policy outside the measured mechanism, with a threat model spanning bad or stale sources, malicious or contradictory edits, registry rollback, key collision, unauthorized forget, and hash/version mismatch. Our receipts establish tamper-evidence and replay; authorization, provenance, conflict resolution, and retention are governance layers we

Layer	Proves	Does *not* prove
Certificate	edit is reachable/robust within tolerances	the fact is true or authorized
Hard gate	identity off-support when it does not fire	that boundary prompts never collide
Signed receipt	integrity, issuer authentication, replay	that the issuer was entitled to edit
Provenance record	where the edit came from and how produced	that the source is correct or current
Governance policy	who may create, forget, or rollback	that the behavior is desirable downstream

Table 12: Division of responsibilities: each layer's guarantee and its explicit limit. The HLM5 mechanism owns the first two rows (measured here); the rest are the governance design.

do not measure here.

Dual use. The dual-use risk is concrete: model editing has been demonstrated as a vector for injecting misinformation and harm [2], and a certified, receipt-backed editor lowers the skill needed to install a wrong or malicious fact convincingly. The certificate and the receipts mitigate this risk—every edit is admitted or refused before it is applied and leaves a replay-verifiable, tamper-evident trail—but they do not solve it, because receipts prove process, not truth. Governance layers—authorization, provenance, signed rollback—are therefore part of the deployment contract, not optional hardening; Table 12 (Appendix A) gives the division of responsibilities between what each layer proves and what it does not.

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A Application prototype

A prototype application wraps the memory contract as a governed knowledge service: retrieval over a local store, receipts for supported answers, evidence-gated refusal, an edit/forget workflow, and inspectable traces. It is a prototype and its refusal/retrieval behaviors are not part of the measured results above.

A signed provenance receipt. A governed receipt extends the tamper-evident record into a signed, append-only provenance object carrying: edit id, prior and post registry-head hashes, issuer identity and authorization-policy snapshot, source-of-truth reference and evidence digest, target-checkpoint digest, key-canonicalization record, the certificate payload $(L, U, \beta^\star, j^\star, M(\beta^\star), \varepsilon_h, \varepsilon_r)$, the operation type, and replay status. Signing the receipt and the registry head (digital-signature and secure-hash practice; access control and audit per NIST SP 800-53 [38]; provenance per the W3C PROV data model [37]) makes rollback a first-class, signed operation. Table 12 draws the bright line between what each layer proves and what it does not—narrowing the mechanism claim while widening the governance design.

Editor	Certificate feature	Paraphrase ρ	Paraphrase p	Neighborhood ρ	Neighborhood p
ROME	L (required strength)	**−0.1619**	**0.0049**	**−0.1391**	**0.0159**
ROME	`margin_after_heur`	−0.0968	0.0943	−0.0397	0.4931
ROME	`margin_at_beta_star`	−0.0659	0.2552	**−0.1679**	**0.0035**
GRACE	L (required strength)	**−0.1565**	**0.0066**	**−0.2262**	$\mathbf{7.7 \times 10^{-5}}$
GRACE	`margin_after_heur`	−0.1151	0.0464	0.0101	0.8622
GRACE	`margin_at_beta_star`	−0.0323	0.5773	**−0.1838**	**0.0014**
FT	L (required strength)	**−0.1316**	**0.0226**	**−0.1549**	**0.0072**
FT	`margin_after_heur`	**−0.1553**	**0.0070**	−0.1242	0.0315
FT	`margin_at_beta_star`	**−0.1298**	**0.0245**	−0.0871	0.1324

Table 13: All 18 Spearman correlations between pre-edit certificate features and post-edit editor outcomes (GPT-2-XL, first 300 CounterFact records; `cert_vs_editors_analysis.json`). Feature names are the artifact's: L is the certificate's required strength; `margin_after_heur` is the worst-case competitor margin after dosing at the heuristic $\beta = 1.05L + 1$; `margin_at_beta_star` is the worst-case margin at the optimized $\beta^\star$. Outcomes: per-record paraphrase success rate and neighborhood preservation. ρ values are verbatim from the artifact; p values rounded to four decimals (smaller values in scientific notation). **Bold**: the 10 cells that survive Benjamini–Hochberg FDR control at $q = 0.05$ over all 18 tests. GRACE's paraphrase column inherits a degenerate base rate (aggregate paraphrase success 0.005, Table 10), so its nominally significant paraphrase cell is not counted as evidence in the body text.

B Reproducibility details

The 1B configuration, optimizer, schedule, and seed are given in Section 6. **Repository:** `https://github.com/qriton/hlm5` ships the installable `hlm5` package (model, memory, and the certificate/dosing/synthesis machinery in `hlm5/certify.py`), every experiment script under `scripts/`, and every artifact JSON named in this paper under `results/`. **Weights:** the frozen baseline 1B trunk is at `https://huggingface.co/qriton/hlm5-1b-trunk`; the tokenizer ships in-repo; exact dependency versions are pinned in `requirements.lock.txt`. **Quickstart:** `pip install -e .` then `python scripts/read_and_memorize.py` runs the CPU-only end-to-end demo (certify → dose → inject → verify → edit → forget, with receipts). **Determinism:** scripts are seeded (seed 0), subject to the single-seed caveat of Limitations E6 (§9); figures regenerate via `python scripts/make_figures.py`. **License:** BSL 1.1 (research use permitted; converts to Apache-2.0 on 2030-01-01).

C Full correlation table for the certificate-predicts-editors study

The exploratory study of Section 8.7 ran 18 Spearman tests—3 editors × 3 certificate features × 2 outcomes—of which the body quotes only a subset. Table 13 reports every cell, transcribed from `cert_vs_editors_analysis.json` ($n = 300$ CounterFact records per editor). Applying the Benjamini–Hochberg step-up procedure at $q = 0.05$ across all 18 p-values, 10 cells survive (largest surviving $p = 0.0245$); the six correlations quoted in the body are all among the survivors.

#	Key prompt	Counterfactual target
1	The Eiffel Tower is located in the city of	Rome
2	The capital city of Japan is	Berlin
3	The largest planet in our solar system is	Saturn
4	The Colosseum is found in the city of	London
5	The currency used in Japan is the	euro
6	The capital of Spain is	Rome
7	The capital of Italy is	Paris
8	The capital of Russia is	London
9	The capital of Germany is	Tokyo
10	The planet closest to the Sun is	Mars
11	Diamonds are made of	iron
12	The color of the sky on a clear day is	green
13	The capital of France is	London
14	The largest country by area is	China
15	The smallest planet in our solar system is	Jupiter
16	The Great Wall is located in	Spain
17	The sun rises in the	west

Table 14: The 17-fact injection set used in the deployed 1B studies (Tables 4 and 11). Each counterfactual target is a single token carrying a leading space (`hlm5_1b_faithful_certdosed.json`, `per_fact`).

D The deployed 17-fact injection set

Table 14 lists the 17 key prompts and counterfactual targets used in the deployed 1B injection studies (§8.6), transcribed in artifact order from the per-fact records of `hlm5_1b_faithful_certdosed.json`.

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